

ORIGINAL RESEARCH

CORONARY

Contemporary Trends, Characteristics, and Outcomes of Patients Undergoing Percutaneous Coronary Intervention for Stent Thrombosis



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ABSTRACT

BACKGROUND There are limited data regarding the temporal trends and outcomes of stent thrombosis (ST) in the contemporary era of potent P2Y₁₂ inhibitors, newer generation drug-eluting stents, and intracoronary imaging.

OBJECTIVES The aim of this study was to examine the trends, characteristics, and outcomes of patients with acute myocardial infarction (AMI) undergoing percutaneous coronary intervention (PCI) for ST in the United States.

METHODS Patients undergoing PCI for AMI between April 2018 and June 2023 were identified from the National Cardiovascular Data Registry CathPCI registry. The rates of PCI for ST were evaluated over time using linear regression. In-hospital outcomes of PCI for ST vs no ST were compared using multivariable logistic regression.

RESULTS Among 2,023,032 patients undergoing PCI for AMI, 57,268 (2.8%) procedures were for ST. The timing of ST was early (18,189 of 55,351 [31.8%]), late (8,825 of 55,351 [15.4%]), very late (28,337 of 55,351 [49.5%]), or unknown (1,917 of 55,351 [3.4%]). There was no temporal change in the frequency of PCI for ST during the study period. PCI for ST was associated with a higher incidence of major adverse cardiac events (adjusted OR [aOR]: 1.24; 95% CI: 1.19-1.29), all-cause mortality (aOR: 1.17; 95% CI: 1.12-1.22), recurrent myocardial infarction (aOR: 1.80; 95% CI: 1.63-1.98), and major bleeding (aOR: 1.18; 95% CI: 1.15-1.22) compared with PCI for non-ST.

CONCLUSIONS In this contemporary, nationwide, real-world analysis, there was no temporal change in the frequency of PCI for ST. Most STs were early or very late. PCI for ST was associated with a higher rate of in-hospital adverse outcomes. (JACC Cardiovasc Interv. 2026;19:541-551) Published by Elsevier on behalf of the American College of Cardiology Foundation.

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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ABBREVIATIONS AND ACRONYMS

ACS = acute coronary syndrome(s)

AMI = acute myocardial infarction

BCIS = British Cardiovascular Intervention Society

LAD = left anterior descending coronary artery

MACE = major adverse cardiac event(s)

MI = myocardial infarction

NCDR = National Cardiovascular Database Registry

NSTEMI = non-ST-segment elevation myocardial infarction

PCI = percutaneous coronary intervention

ST = stent thrombosis

STEMI = ST-segment elevation myocardial infarction

Stent thrombosis (ST) is a rare but serious complication of percutaneous coronary intervention (PCI) and often presents as acute myocardial infarction (AMI).^{1,2} ST can be classified according to time from the index procedure as early (up to 30 days), late (31-365 days), and very late (>365 days).^{1,2} The pathophysiological mechanism for ST differs according to the timing. Early ST is likely due to technical or procedural factors or a lack of adequate platelet activation inhibition, whereas late and very late ST is likely due to nonatherosclerotic plaques, minimal endothelialization, and/or late stent malapposition.¹⁻³

A prior analysis of the National Cardiovascular Database Registry's (NCDR) CathPCI database registry included angiographically confirmed ST from 2009 to 2010 and showed that the prevalence of ST was 1.8% among patients with acute coronary syndrome (ACS).⁴ In that analysis, very late was the predominant type of ST (61%).⁴ During the past decade, PCI has undergone several major advances, including the widespread use of potent P2Y₁₂ inhibitors, newer generation drug-eluting stents, and the increased use of intracoronary imaging to optimize PCI results.⁵⁻⁸ It is unknown whether there has been a reduction in the incidence of ST and whether there has been a change in the factors and timing of ST events. We examined contemporary trends, clinical characteristics, and in-hospital outcomes among patients with AMI undergoing PCI for ST from the NCDR CathPCI registry.

METHODS

DATA SOURCE AND STUDY POPULATION. The NCDR CathPCI registry is a voluntary registry that captures PCI care at >2,000 U.S. hospitals using a standardized data collection. The data collection process is monitored by a data quality committee, and annual audits are performed to ensure data accuracy. Details regarding data abstraction and variable definitions have been published.⁹ This analysis was conducted using version 5.0 of the NCDR CathPCI registry. Institutional Review Board approval was obtained from Duke University Medical Center, which determined that the study did not require informed consent.

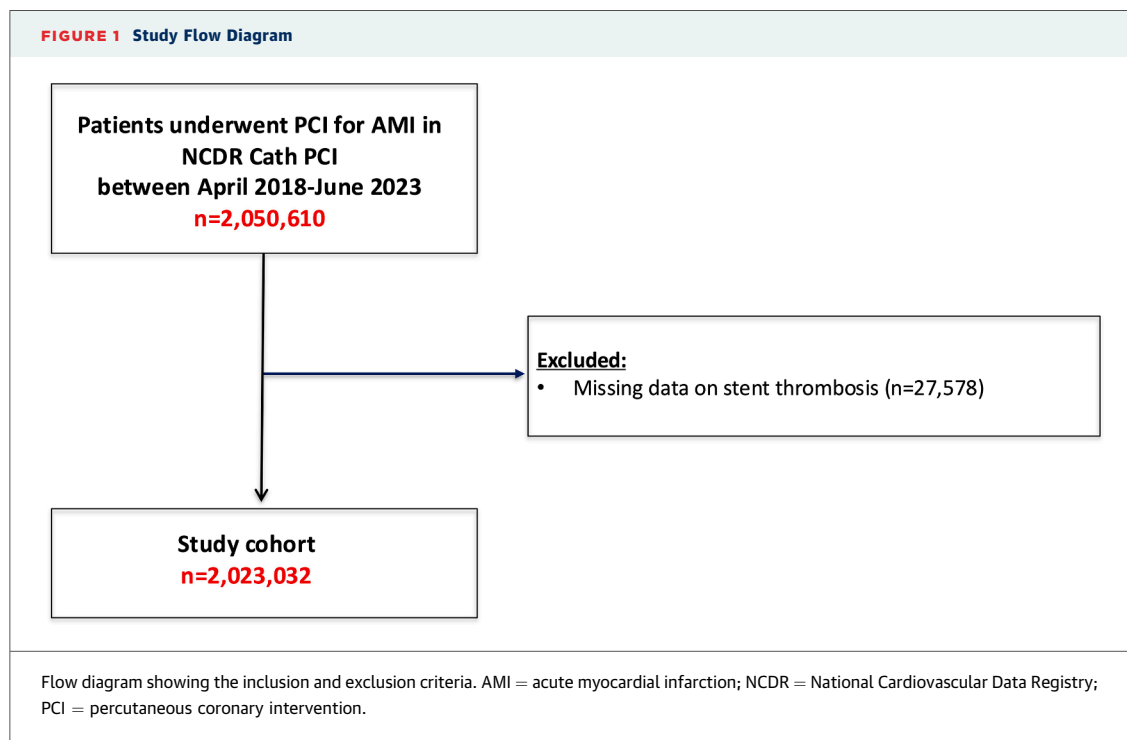
Patients who underwent PCI for AMI (defined as ST-segment elevation myocardial infarction [STEMI]

or non-ST-segment elevation myocardial infarction [NSTEMI]) at any participating site from April 2018 through June 2023 were included. Patients with missing data on ST status were excluded.

STUDY OUTCOMES. The primary outcome was the incidence of PCI for ST. ST was stratified according to the time from the index procedure as follows: early (up to 30 days), late (31-365 days), and very late (>365 days).^{5,6} The secondary outcomes included in-hospital major adverse cardiac events (MACE; defined as the composite of all-cause mortality, myocardial infarction [MI] or stroke), all-cause mortality, MI, stroke, and NCDR major bleeding (defined as any in-hospital bleeding, in-hospital cardiac tamponade, in-hospital hemorrhagic stroke, requirement of blood transfusion with preprocedural hemoglobin >8 g/dL excluding post-coronary artery bypass surgery patients, or a >3 g/dL difference between pre- and postprocedural hemoglobin level among patients with preprocedural hemoglobin ≤16 g/dL).^{9,10}

STATISTICAL ANALYSIS. The baseline patient, procedural, and hospital-level characteristics were compared between ST and no ST and for the timing of ST (early, late, and very late). The baseline characteristics and definitions of variables are provided at the NCDR CathPCI website (<https://cvquality.acc.org/NCDR-Home/registries/hospital-registries/cathpci-registry>). Categorical variables are presented as frequencies and were assessed using the Pearson chi-square test, while continuous variables are presented as median (Q1-Q3) and were compared using the Wilcoxon test. Linear regression analysis accounting for autocorrelation and the nonparametric Mann-Kendall test were used to test for the temporal trends.

We constructed a multivariable logistic regression model to assess the association between ST and in-hospital outcomes. The regression model was adjusted for the patient, procedural, and hospital-level variables that have been shown to be associated with worse outcomes among AMI patients. The following variables were included: age, sex, body mass index, current smoking, hypertension, hyperlipidemia, diabetes mellitus, prior MI, prior cerebrovascular disease, prior peripheral arterial disease, prior renal failure (ie, glomerular filtration rate <30 mL/min/1.73 m² or on dialysis), prior heart failure, prior coronary artery bypass surgery, presentation (ie, STEMI vs NSTEMI), left anterior descending coronary artery (LAD) culprit (vs non-LAD culprit), preintervention TIMI flow grade of the culprit lesion,



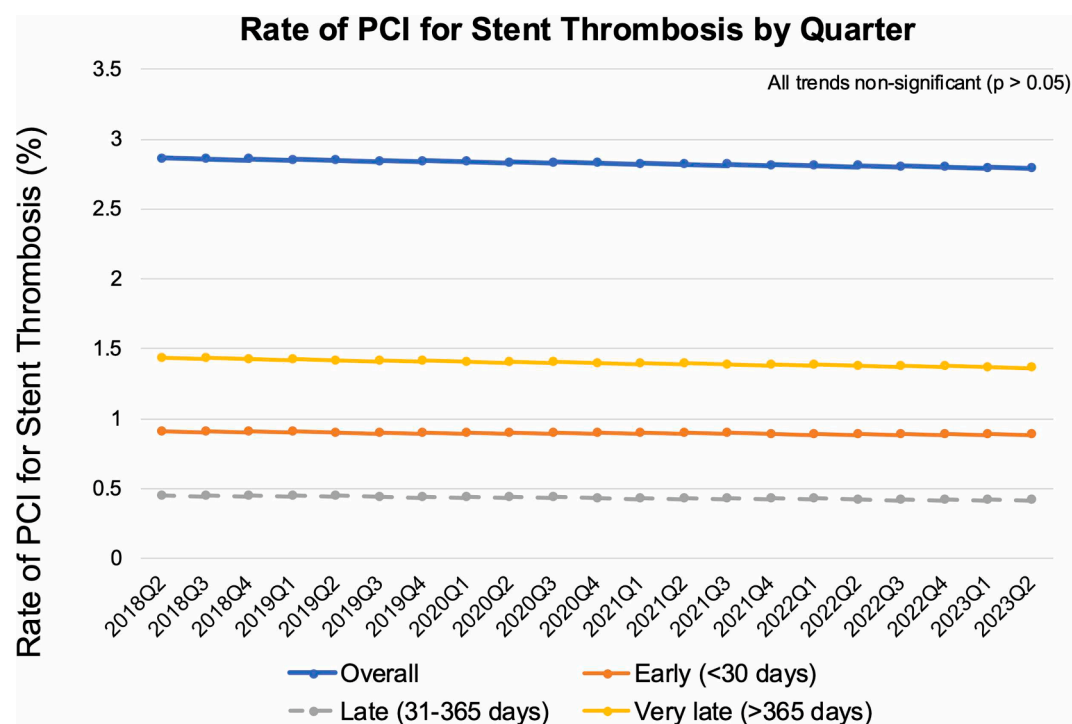
preprocedural creatinine, P2Y₁₂ inhibitor (ie, clopidogrel vs high-potency P2Y₁₂ inhibitor), use of unfractionated heparin (vs bivalirudin), use of glycoprotein IIb/IIIa inhibitors, cardiac arrest, cardiogenic shock, radial access (vs femoral access), lesion complexity, bifurcation lesion, thrombus aspiration, hospital community type, hospital profit type, and teaching hospital status. Because of minimal missingness in the adjusted variables (<0.4%), imputation was not performed. To account for socioeconomic status, a sensitivity analysis was accounted using the same approach for admissions with nonmissing patient zip code (80% of the overall analytical data set) (model 2). In this sensitivity analysis, socioeconomic status was accounted for by adding the following variables to the model: median household income, median home value, high school and college graduate rate, and unemployment rate from the 2021 U.S. census social economic data. To identify factors independently associated with MACE among patients with ST, we conducted a second multivariable logistic regression including the following variables: age, sex, prior cerebrovascular disease, prior peripheral arterial disease, diabetes mellitus, chronic kidney disease, presentation (ie, STEMI vs NSTEMI), LAD

culprit (vs non-LAD culprit), cardiac arrest, cardiogenic shock, radial access (vs femoral access), number of stents, and bifurcation lesions. These variables were selected on the basis of previous studies.^{4,11} Variables with *P* values <0.05 in the final model were retained. For each model, we report adjusted ORs, 95% CIs, and *P* values on the basis of the final set of selected variables. All statistical analyses were performed using SAS version 9.4 (SAS Institute).

RESULTS

INCIDENCE AND TEMPORAL TRENDS FOR PCI AND

ST. Of the 2,050,610 PCIs performed for AMI from April 2018 to June 2023, 27,578 (1.34%) with missing ST status or missing date of previously treated lesion were excluded (Figure 1). Of the remaining 2,023,032 patients, 57,268 (2.83%) had ST. According to the timing, ST was early (18,189 of 55,351 [31.8%]), late (8,825 of 55,351 [15.4%]), very late (28,337 of 55,351 [49.5%]), or unknown (1,917 of 55,351 [3.4%]). There was no change over time during the study period in the overall incidence for PCI for ST (*P*_{trend} = 0.21), nor when stratified by timing of ST (Figure 2).

FIGURE 2 Scatterplot for the Temporal Trends in PCI for Stent Thrombosis, Stratified by Timing

PCI = percutaneous coronary intervention.

BASELINE CHARACTERISTICS. Compared with patients without ST, those who underwent PCI for ST were younger (64.3 vs 65.4 years; $P < 0.001$) and more likely to be men. Patients with ST were more likely to be current smokers and had a higher prevalence of hypertension, diabetes mellitus, prior MI, prior cerebrovascular disease, peripheral vascular disease, chronic lung disease, and prior coronary artery bypass surgery. Patients with ST were more likely to present with STEMI, cardiac arrest, cardiogenic shock, ventricular arrhythmias (ventricular fibrillation or tachycardia), or high-grade heart block (second- or third-degree heart block) and to have a left ventricular ejection fraction $<40\%$. [Table 1](#) summarizes the patient and hospital characteristics among those who underwent PCI for ST vs those who underwent PCI for non-ST.

[Table 2](#) presents the patient and hospital characteristics of early, late, and very late ST. The proportion of women was higher among patients presenting with early ST compared with late and very late ST. Patients presenting with early ST were less likely to be current smokers and had a lower prevalence of hypertension, hyperlipidemia, and prior MI, but a

higher likelihood of having diabetes mellitus, compared with patients with very late ST. Patients with early ST were also more likely to present with cardiogenic shock, cardiac arrest, ventricular arrhythmia, or complete heart block compared with those with late and very late ST.

PROCEDURAL CHARACTERISTICS. Compared with patients without ST, femoral access was more frequently used among patients with ST, and the culprit vessel among patients with ST was more likely to be the LAD. Intravenous glycoprotein IIb/IIIa inhibitors, cangrelor, and potent P2Y₁₂ inhibitors (prasugrel or ticagrelor) were also more likely to be used among patients undergoing PCI for ST compared with those without ST. The use of thrombus aspiration devices was higher among patients with ST ([Table 1](#)). [Table 2](#) summarizes the procedural characteristics for patients with early, late, and very late ST. The culprit lesion was more likely LAD among the early ST group. The use of ventricular support devices was more frequent in the early ST group.

IN-HOSPITAL OUTCOMES. There was an association between ST and higher incidence of in-hospital

TABLE 1 Baseline Patient, Hospital-Level, and Procedural Characteristics Among Patients With Acute Myocardial Infarction Undergoing Percutaneous Coronary Intervention

	Stent Thrombosis (n = 57,268)	No Stent Thrombosis (n = 1,965,764)	P Value
Age, y	64 (56-73)	66 (57-74)	<0.0001
BMI, kg/m ²	29 (25.5-33.1)	29 (25.6-33.4)	<0.0001
Female	14,866 (26.0)	626,739 (31.9)	<0.0001
Race			
White	47,519 (83.0)	1,634,833 (83.2)	0.23
Black	6,328 (11.0)	187,198 (9.5)	<0.0001
Asian	1,302 (2.3)	65,485 (3.3)	<0.0001
American Indian/Alaskan Native	458 (0.8)	13,397 (0.7)	0.0007
Native Hawaiian/Pacific Islander	144 (0.2)	5,719 (0.3)	0.08
Current smoking	18,921 (33.1)	519,025 (26.4)	<0.0001
Hypertension	50,690 (88.5)	1,577,817 (80.3)	<0.0001
Diabetes mellitus	24,178 (42.2)	778,943 (39.6)	<0.0001
Hyperlipidemia	49,727 (86.9)	1,416,226 (72.1)	<0.0001
Prior MI	38,612 (67.4)	483,208 (24.6)	<0.0001
Prior PCI	57,268 (100.0)	622,388 (31.7)	<0.0001
Previous left main PCI	1,389 (2.9)	18,192 (3.2)	0.001
Cerebrovascular disease	9,005 (15.7)	265,402 (13.5)	<0.0001
Renal failure (GFR <30 mL/min/1.73 m ² or dialysis)	3,722 (6.5)	127,478 (6.5)	0.89
Prior heart failure	16,953 (29.6)	475,702 (24.2)	<0.0001
Prior PAD	7,117 (12.4)	193,635 (9.9)	<0.0001
Prior CABG	7,921 (13.8)	248,994 (12.7)	<0.0001
STEMI	36,926 (64.5)	599,750 (30.5)	<0.0001
NSTEMI	20,342 (35.5)	1,366,014 (69.5)	<0.0001
LVEF <40%	7,241 (26.6)	190,962 (20.1)	<0.0001
Cardiogenic shock	5,629 (14.4)	86,492 (11.3)	<0.0001
Out-of-hospital cardiac arrest	1,607 (2.8)	44,523 (2.3)	<0.0001
Hospital community			
Rural	9,366 (16.3)	299,169 (15.2)	<0.0001
Suburban	18,354 (32.0)	669,763 (34.1)	
Urban	29,547 (51.6)	996,771 (50.7)	
Teaching hospital	28,015 (48.9)	999,251 (50.8)	
Procedure related			
Culprit vessel			
LAD	23,607 (41.2)	746,626 (38.0)	<0.0001
RCA	21,428 (37.4)	614,226 (31.2)	<0.0001
LCx	10,821 (18.9)	449,796 (22.9)	<0.0001
Bypass graft	2,770 (4.8)	75,407 (3.8)	<0.0001
Preprocedural TIMI flow grade			
0	37,399 (70.1)	497,562 (29.2)	<0.0001
1 or 2	8,737 (16.4)	467,969 (27.5)	
3	7,177 (13.5)	737,863 (43.3)	
Culprit lesion TIMI flow grade postintervention			
3	49,574 (94.6)	1,630,618 (97.1)	<0.0001
P2Y ₁₂ inhibitor during procedure			
Clopidogrel	18,316 (32.0)	783,952 (39.9)	<0.0001
Ticagrelor	30,901 (54.0)	866,876 (44.1)	<0.0001
Prasugrel	5,942 (10.4)	169,866 (8.6)	<0.0001
Cangrelor	3,080 (5.4)	58,520 (3.0)	<0.0001

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TABLE 1 Continued

	Stent Thrombosis (n = 57,268)	No Stent Thrombosis (n = 1,965,764)	P Value
Anticoagulant agent used during procedure			
Unfractionated heparin	49,868 (87.1)	1,716,653 (87.4)	0.10
Low-molecular weight heparin	2,650 (4.6)	149,259 (7.6)	<0.0001
Glycoprotein IIb/IIIa inhibitors	24,996 (43.7)	333,163 (17.0)	<0.0001
Bivalirudin	12,132 (21.2)	415,391 (21.1)	<0.0001
Use of mechanical circulatory support	5,473 (9.6)	101,673 (5.2)	0.01
Access site			
Radial	23,739 (41.5)	1,050,828 (53.5)	<0.0001
Femoral	33,277 (58.1)	907,658 (46.2)	<0.0001
High-complexity lesion	39,838 (69.8)	1,322,010 (67.5)	<0.0001
Bifurcation lesion	7,555 (13.2)	233,976 (11.9)	<0.0001
Multivessel PCI	20,529 (35.9)	778,751 (39.6)	<0.0001
Number of stents	1.00 (0.0-1.00)	1.00 (1.00-2.00)	<0.0001
Thrombus aspiration	17,477 (30.5)	124,370 (6.3)	<0.0001
P2Y ₁₂ inhibitor upon discharge			
Clopidogrel	17,130 (34.9)	932,604 (51.0)	<0.0001
Ticagrelor	7,028 (14.1)	175,608 (9.6)	<0.0001
Prasugrel	25,711 (52.0)	710,272 (38.9)	<0.0001
Oral anticoagulant agent upon discharge	7,600 (15.1)	230,809 (12.5)	<0.0001

Values are median (Q1-Q3).

BMI = body mass index; CABG = coronary artery bypass grafting; LVEF = left ventricular ejection fraction; GFR = glomerular filtration rate; LAD = left anterior descending coronary artery; LCx = left circumflex coronary artery; MI = myocardial infarction; NSTEMI = non-ST-segment elevation myocardial infarction; RCA = right coronary artery; = peripheral artery disease; PCI = percutaneous coronary intervention; STEMI = ST-segment elevation acute myocardial infarction.

MACE (7.1% [95% CI: 6.8-7.3] vs 3.5% [95% CI: 3.47-3.52]; OR: 1.24; 95% CI: 1.19-1.29; $P < 0.001$), driven by higher mortality (6.3% [95% CI: 6.1-6.5] vs 3.2% [95% CI: 3.16-3.21]; OR: 1.17; 95% CI: 1.12-1.22; $P < 0.001$), recurrent MI (0.9% [95% CI: 0.86-1.02] vs 0.38% [95% CI: 0.37-0.39]; OR: 1.80; 95% CI: 1.63-1.98; $P < 0.001$), and major bleeding (12.2% [95% CI: 11.9-12.4] vs 6.5% [6.47-6.54]; OR: 1.18; 95% CI: 1.15-1.22; $P < 0.001$), while there was no difference in stroke compared with patients without ST (**Central Illustration**). The ST group had a longer median hospital stay (3 days [Q1-Q3: 2-4 days] vs 2 days [Q1-Q3: 2-4 days]; $P < 0.001$). The findings were consistent on sensitivity analyses accounting for patient socioeconomic status. **Table 3** presents the unadjusted and adjusted in-hospital outcomes for the ST vs none.

FACTORS INDEPENDENTLY ASSOCIATED WITH MACE AMONG ST GROUP. On multivariable analyses, the following factors were independently associated with higher MACE among patients with ST: older age (>75 years) (OR: 1.21; 95% CI: 1.16-1.25), diabetes mellitus (OR: 1.34; 95% CI: 1.25-1.44), prior cerebrovascular disease (OR: 1.20; 95% CI: 1.10-1.32), peripheral arterial disease (OR: 1.30; 95% CI: 1.18-1.44), chronic kidney disease (OR: 2.07; 95% CI: 1.86-2.31), STEMI presentation (OR: 1.43; 95% CI: 1.31-1.55), cardiac arrest (OR: 3.31; 95% CI: 2.94-3.74), cardiogenic

shock (OR: 8.43; 95% CI: 7.81-9.11), LAD culprit lesion (OR: 1.61; 95% CI: 1.49-1.74), and femoral access compared with radial (OR: 1.62; 95% CI: 1.49-1.76), while male sex was associated with a lower incidence of MACE (OR: 0.80; 95% CI: 0.74-0.86) (**Table 4**).

DISCUSSION

In this large-scale observational national registry of about 2 million all-comers with AMI undergoing PCI from April 2018 to June 2023, we evaluated the temporal trends and outcomes of ST in the contemporary era. The main study findings were as follows: 1) the frequency of ST was 2.8%, and almost one-half of the cases were due to very late ST; 2) patients with ST were younger and more likely to be men, active smokers, had multiple comorbidities, and presented with STEMI, cardiac arrest, cardiogenic shock, and arrhythmias; 3) there were no significant temporal changes in the incidence of PCI for ST; 4) patients with ST had a higher adjusted risk for MACE driven by a higher incidence of in-hospital mortality and recurrent MI, as well as higher rates of major bleeding; and 5) the following factors were associated with higher MACE among patients with ST: older age, diabetes, prior cerebrovascular or peripheral arterial disease, chronic kidney disease, prior heart failure as well as STEMI presentation, cardiac arrest, and LAD

TABLE 2 Baseline Patient, Hospital-Level, Procedural Characteristics According to Timing of ST

	Early ST (0-30 d) (n = 18,189)	Late ST (31-365 d) (n = 8,825)	Very Late ST (>365 d) (n = 28,337)	P Value
Age, y	64 (56-72)	64 (55-73)	64 (56-73)	<0.0001
BMI, kg/m ²	29 (25.4-33.2)	29.2 (25.4-33.2)	29 (25.5-33.0)	0.08
Females	5,826 (32.0)	2,402 (27.2)	6,177 (21.8)	<0.0001
Race				
White	14,819 (81.4)	7,043 (79.8)	24,100 (85.0)	<0.0001
Black	2,179 (12.0)	1,295 (14.7)	2,622 (9.2)	<0.0001
Asian	448 (2.5)	201 (2.3)	630 (2.2)	0.24
American Indian/Alaskan Native	171 (0.9)	78 (0.9)	197 (0.7)	0.01
Native Hawaiian/Pacific Islander	54 (0.3)	25 (0.3)	56 (0.2)	0.07
Current smoking	5,559 (30.6)	2,741 (31.1)	9,969 (35.2)	<0.0001
Hypertension	15,491 (85.2)	8,161 (92.5)	25,313 (89.4)	<0.0001
Diabetes mellitus	8234 (45.3)	4,566 (51.7)	10,567 (37.3)	<0.0001
Hyperlipidemia	14,793 (81.3)	7,950 (90.1)	25,338 (89.4)	<0.0001
Prior MI	10,257 (56.4)	6,702 (75.9)	20,481 (72.3)	<0.0001
Previous left main PCI	384 (3.1)	470 (5.6)	498 (2.0)	<0.0001
Cerebrovascular disease	2,927 (16.1)	1,705 (19.3)	4,033 (14.2)	<0.0001
Renal failure (GFR <30 mL/min/1.73 m ² or dialysis)	926 (5.1)	1,351 (15.3)	1,301 (4.6)	<0.0001
Prior heart failure	5,700 (31.4)	3,556 (40.3)	7,133 (25.2)	<0.0001
Prior PAD	2,086 (11.5)	1,586 (18.0)	3,202 (11.3)	<0.0001
Prior CABG	1,916 (10.5)	1,820 (20.6)	3,807 (13.4)	<0.0001
STEMI	12,445 (68.4)	4,429 (50.2)	18,772 (66.2)	<0.0001
NSTEMI	5,744 (31.6)	4,396 (49.8)	9,565 (33.7)	<0.0001
LVEF <40%	3,650 (27.5)	1,462 (28.3)	1,955 (23.8)	<0.0001
Cardiogenic shock	2,342 (16.8)	735 (14.7)	2,309 (12.3)	<0.0001
Out-of-hospital cardiac arrest	538 (3.0)	176 (2.0)	812 (2.9)	<0.0001
Hospital community				0.04
Rural	2,895 (15.9)	1,455 (16.5)	4,729 (16.7)	
Suburban	5,908 (32.5)	2,744 (31.1)	9,114 (32.2)	
Urban	9,386 (51.6)	4,626 (52.4)	14,494 (51.1)	
Teaching hospital	8,708 (47.9)	4,299 (48.7)	14,080 (49.7)	0.0006
Procedure related				
Culprit vessel				
LAD	8,723 (48.0)	3,431 (38.9)	10,790 (38.1)	<0.0001
RCA	5,615 (30.9)	3,051 (34.6)	12,007 (42.4)	<0.0001
LCx	3,604 (19.8)	2,005 (22.7)	4,840 (17.1)	<0.0001
Bypass graft	387 (2.1)	639 (7.2)	1,609 (5.7)	<0.0001
Preprocedural TIMI flow grade				
0	13,694 (80.5)	4,666 (57.7)	17,870 (67.4)	<0.0001
1 or 2	1,643 (9.7)	1,870 (23.1)	4,907 (18.5)	<0.0001
3	1,674 (9.8)	1,552 (19.2)	3,737 (14.1)	<0.0001
Culprit lesion TIMI flow grade postintervention				
3	15,493 (93.3)	7,508 (94.7)	25,007 (95.4)	<0.0001
P2Y ₁₂ inhibitor during procedure				
Clopidogrel	5,802 (31.9)	3,274 (37.1)	8,585 (30.3)	<0.0001
Ticagrelor	10,303 (56.7)	4,429 (50.2)	15,244 (53.8)	<0.0001
Prasugrel	2,033 (11.2)	982 (11.1)	2,766 (9.8)	<0.0001
Anticoagulant agent used during procedure				
Unfractionated heparin	78 (0.9)	55 (1.2)	111 (0.8)	0.02
Low-molecular weight heparin	816 (4.5)	452 (5.1)	1,296 (4.6)	0.05
Glycoprotein IIb/IIIa inhibitors	10,383 (57.1)	3,239 (36.7)	10,655 (37.6)	<0.0001
Bivalirudin	4,085 (22.5)	1,857 (21.1)	5,779 (20.4)	<0.0001

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culprit lesions, while radial access was associated with lower MACE risk.

The present analysis represents the largest contemporary cohort characterizing patients

undergoing PCI for ST. Among patients presenting with AMI, those with ST had distinct demographics and clinical profile; they were more often younger men with a higher prevalence of risk factors and

TABLE 2 Continued

	Early ST (0-30 d) (n = 18,189)	Late ST (31-365 d) (n = 8,825)	Very Late ST (>365 d) (n = 28,337)	P Value
Use of mechanical circulatory support	2,493 (62.3)	783 (57.4)	2,008 (48.1)	<0.0001
Access site				
Radial	5,955 (32.7)	3,240 (36.7)	13,784 (48.6)	<0.0001
Femoral	12,166 (66.9)	5,533 (62.7)	14,433 (50.9)	<0.0001
High-complexity lesion	11,497 (63.5)	5,717 (65.1)	21,235 (75.1)	<0.0001
Bifurcation lesion	2,397 (13.2)	1,260 (14.3)	3,678 (13.0)	0.006
Multivessel PCI	5,531 (30.4)	3,185 (36.1)	10,958 (38.7)	<0.0001
Number of stents	1.00 (0.0-1.00)	1.00 (0.0-1.00)	1.00 (1.00-2.00)	<0.0001
Thrombus aspiration	6,417 (35.3)	2,334 (26.4)	8,175 (28.8)	<0.0001
P2Y ₁₂ inhibitor upon discharge				
Clopidogrel	3,724 (25.2)	2,830 (37.4)	9,907 (39.4)	<0.0001
Ticagrelor	2,506 (16.5)	1,181 (15.3)	3,152 (12.4)	<0.0001
Prasugrel	8,982 (59.7)	3,701 (48.6)	12,278 (48.7)	<0.0001
Oral anticoagulant upon discharge	2,591 (16.9)	1,435 (18.4)	3,308 (12.9)	<0.0001

Values are median (Q1-Q3) or n (%).

ST = stent thrombosis; other abbreviations as in Table 1.

comorbid conditions. Furthermore, patients with ST were more likely to present with STEMI and other high-acuity findings, such as cardiogenic shock, cardiac arrest, or significant arrhythmias. The clinical and demographic profile in this analysis is in line with data from the British Cardiovascular Intervention Society (BCIS) registry.¹¹ It is important to note that this analysis spanned during the COVID-19 pandemic; COVID-19 is a well-recognized risk factor for ST.^{12,13} Nevertheless, our analysis showed that the rates of ST remained mostly stable even after 2022 and highlights the challenging nature for patients with ST, who constitute a high-risk group.

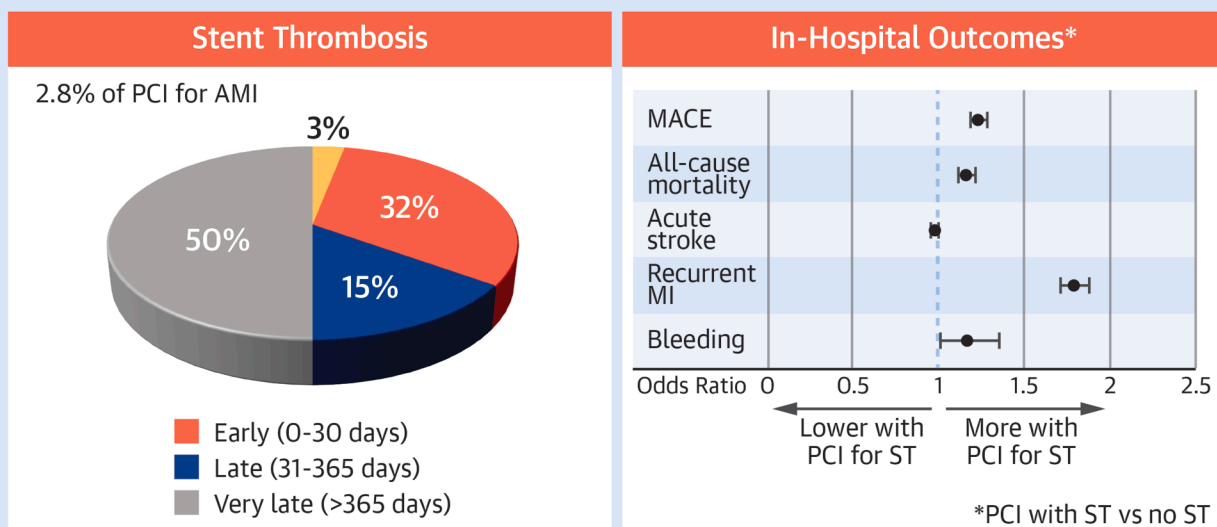
In this analysis, we observed differences in demographics and clinical profile according to the timing of ST. Patients with early and late ST had a higher likelihood of being women and Black and had a higher rate of diabetes. This is in line with earlier data from the CathPCI registry (2009-2010) and data from BCIS registry.^{4,11} These differences might be related to differences in pathophysiology, as suggested by prior studies. Early ST is likely due to stent underexpansion, acute malapposition, and premature discontinuation of antiplatelet therapy, particularly before reendothelialization has occurred. Unfortunately, data on antiplatelet duration and discontinuation are not captured in the CathPCI registry. By contrast, very late ST has been previously linked to late stent malapposition, delayed endo-endothelialization, and the development of neoatherosclerosis.¹⁴

Very late ST (49%) was most common, followed by early and by late ST in this study. This is consistent with the earlier analysis from the CathPCI registry (2009-2010), which included predominately

first-generation drug-eluting stents, in which very late ST was also the most common type (61%).⁴ In contrast to our study, an analysis from the 2014 to 2020 BCIS involving any PCI (not restricted to ACS) showed that early ST was most common (52.6%), followed by very late ST (35.4%).¹¹ The variations in the design of these registries limit definite conclusions behind differences in reported timing of ST. It is plausible that regional factors may influence the timing of ST, including variations in patient characteristics, procedural approaches, and antithrombotic strategies. The patient population in our cohort had a higher burden of comorbidities, including diabetes, renal failure, peripheral vascular disease, and prior cerebrovascular disease, which might contribute to higher risk for late stent-related complications such as neoatherosclerosis. The higher early ST rate in the BCIS registry might reflect significant differences in patient or procedural characteristics during the index PCI admission, such as a greater proportion of PCI for ACS, differences in procedural stent optimization, or the use of intracoronary imaging, which increase the risk for early thrombotic events. In the BCIS analysis, there was a decline in the incidence of stent thrombosis per 100,000 population, which was notable mainly between 2014 and 2018, while rates did not decline between 2018 and 2020.¹¹ Our contemporary analysis between 2018 and 2023 demonstrated no significant change in the rates of ST as a proportion of all admissions with AMI. Technical and patient factors contribute to persistent rates of ST. Details regarding the index PCI procedure (eg, bifurcation technique, degree of calcifications, intravascular imaging use) are not available and could partly explain these persistently high rates of ST.

CENTRAL ILLUSTRATION Rates and Outcomes of Stent Thrombosis in the NCDR CathPCI Registry

Rates of Stent Thrombosis (ST) and Outcomes of Patients Undergoing PCI for AMI From 2018-2023 in the NCDR Cath PCI Registry N = 2,023,032



- In this contemporary, nationwide, real-world analysis, there was no temporal change in the frequency of PCI for ST.
- Most ST were early (up to 30 days; 32%) or very late (>365 days; 50%).
- PCI for ST was associated with higher in-hospital adverse outcomes.

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(Left) Frequency of stent thrombosis, early stent thrombosis (0-30 days), late stent thrombosis (31-365 days), and very late stent thrombosis (>365 days). (Right) Forest plot for adjusted in-hospital outcomes for percutaneous coronary intervention (PCI) for stent thrombosis (ST) vs no stent thrombosis. AMI = acute myocardial infarction; MACE = major adverse cardiac event(s); MI = myocardial infarction; NCDR = National Cardiovascular Data Registry.

TABLE 3 In-Hospital Outcomes After Percutaneous Coronary Intervention for Stent Thrombosis vs No Stent Thrombosis

	Stent Thrombosis/None Incidence, % (95% CI)	Unadjusted		Adjusted Model 1		Adjusted Model 2	
		OR (95% CI)	P Value	OR (95% CI)	P Value	OR (95% CI)	P Value
Major adverse cardiac events	7.1 (6.8-7.3)/3.5 (3.47-3.52)	2.10 (2.03-2.17)	<0.001	1.24 (1.19-1.29)	<0.001	1.27 (1.22-1.33)	<0.001
All-cause mortality	6.3 (6.1-6.5)/3.2 (3.16-3.21)	2.05 (1.98-2.12)	<0.001	1.17 (1.12-1.22)	<0.001	1.20 (1.14-1.26)	<0.001
Acute stroke	0.9 (0.80-0.95)/0.6 (0.59-0.61)	1.46 (1.33-1.59)	<0.001	0.99 (0.90-1.08)	0.77	1.01 (0.91-1.13)	0.81
Recurrent myocardial infarction	0.9 (0.86-1.02)/0.38 (0.37-0.39)	2.48 (2.27-2.70)	<0.001	1.80 (1.63-1.98)	<0.001	1.89 (1.70-2.11)	<0.001
Major bleeding	12.2 (11.9-12.4)/6.5 (6.47-6.54)	1.99 (1.94-2.04)	<0.001	1.18 (1.15-1.22)	<0.001	1.19 (1.15-1.23)	<0.001

Model 1 was adjusted for patient, procedural, and hospital-related factors. Model 2 was adjusted for socioeconomic status in addition to patient, procedural, and hospital-related factors.

TABLE 4 Multivariate Analysis of Factors Associated With In-Hospital Major Adverse Cardiac Events Among Patients Who Underwent Percutaneous Coronary Intervention for Stent Thrombosis

	OR (95% CI)	P Value
Age >75 y, each 5-y increase	1.21 (1.16-1.25)	<0.0001
Male	0.80 (0.74-0.86)	<0.0001
Diabetes mellitus	1.34 (1.25-1.44)	<0.0001
Prior cerebrovascular disease	1.20 (1.10-1.32)	<0.0001
Peripheral arterial disease	1.30 (1.18-1.44)	<0.0001
Chronic kidney disease	2.07 (1.86-2.31)	<0.0001
ST-segment elevation myocardial infarction	1.43 (1.31-1.55)	<0.0001
Cardiac arrest	3.31 (2.94-3.74)	<0.0001
Cardiogenic shock	8.43 (7.81-9.11)	<0.0001
LAD vs non-LAD vessel	1.61 (1.49-1.74)	<0.0001
Femoral arterial access site (vs radial)	1.62 (1.49-1.76)	<0.0001
Number of stents	1.00 (0.96-1.03)	0.82
Bifurcation lesion	0.99 (0.89-1.1)	0.82

LAD = left anterior descending coronary artery.

We found that PCI for ST was associated with higher risk for MACE, as well as mortality, recurrent MI and major bleeding. These results confirm the previously reported worse outcomes with ST in older studies. The earlier analysis from the CathPCI registry (2009-2010) and the analysis from the BCIS registry both showed similar higher adverse cardiac events with PCI for ST.¹¹ The worse outcomes observed with ST are likely multifactorial. The pronounced thrombotic milieu in patients with ST may explain the higher risk for cardiogenic shock, arrest, and unstable arrhythmias at presentation. The higher risk for MACE was also mediated by a higher risk for recurrent MI and bleeding among patients with ST, even after adjusting for demographics, and clinical presentation. This highlights the critical challenges in the management of patients with ST, such as achieving a balance between a potent antithrombotic regimen and bleeding risk. Although there have been technical advances in procedural and pharmacologic approaches for treatment of heavy coronary thrombus burden, our contemporary analysis demonstrated persistently worse outcomes with PCI for ST. Multivariable analysis did not demonstrate a significant benefit for using thrombus aspiration devices among patients with ST, which aligns with the lack of benefit of routine thrombus aspiration among all-comers with STEMI in large-scale randomized trials.^{15,16} We also demonstrated that strongest factors associated with MACE among patients with ST included cardiogenic shock, cardiac arrest, chronic kidney disease, age > 75 years, female sex, femoral

access, STEMI presentation, LAD culprit, and peripheral arterial disease. These results are aligned with those of the BCIS registry.¹¹ This underscores the complexity of managing ST in patients with multiple comorbidities. The protective effect of radial access is likely related to the reduced major bleeding and mortality among patients with ST, who require more potent and prolonged antithrombotic regimens.¹⁷

STUDY LIMITATIONS. First, the observational nature of this study poses a possibility for residual confounding. We performed robust multivariable regression analysis to reduce the risk for allocation bias.

Second, certain data were not available in the present data set, such as data on the index stent implantation procedure, including stent length, diameter, expansion, malposition, use of intracoronary imaging, and antiplatelet therapies. Information regarding medication compliance, particularly antiplatelet agents, which play an important role in the pathophysiology of early ST were also not available.

Third, participation in the CathPCI registry is voluntary, and the participating sites are usually high-performing sites, which introduces a risk for selection bias. Last, long-term outcomes data were not available in the current data set.

CONCLUSIONS

In this large-scale, nationwide, contemporary observational analysis of all-comers undergoing PCI for AMI, we found that the rate of ST was about 2.8%, and early and very late ST compromised almost one-third and one-half of all ST cases, respectively. There was no temporal change in the incidence of ST. ST patients were younger and had high-risk features on presentation. PCI for ST was associated with a higher incidence of MACE, in-hospital mortality, recurrent MI, and major bleeding. Additional efforts should focus on identifying the factors contributing to the persistent rates of ST in the contemporary PCI era.

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PERSPECTIVES

WHAT IS KNOWN? There is a paucity of contemporary data regarding the temporal trends and outcomes of ST considering the widespread use of potent P2Y₁₂ inhibitors, newer generation drug-eluting stents, and intra-coronary imaging.

WHAT IS NEW? Among patients undergoing PCI for AMI, the rate of ST was 2.8%, and almost one-half of the cases were due to very late ST. Patients with ST were younger and more likely to be men and to present with STEMI, cardiac arrest, cardiogenic shock, and

arrhythmias. There were no significant temporal changes in the incidence of PCI for ST. Patients with ST had a higher adjusted incidence of MACE, driven by a higher incidence of in-hospital mortality and recurrent MI, as well as higher rates of major bleeding.

WHAT IS NEXT? Additional efforts should focus on identifying the factors contributing to the persistent rates of ST in the current PCI era and optimizing ST treatment, such as using intravascular imaging.

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